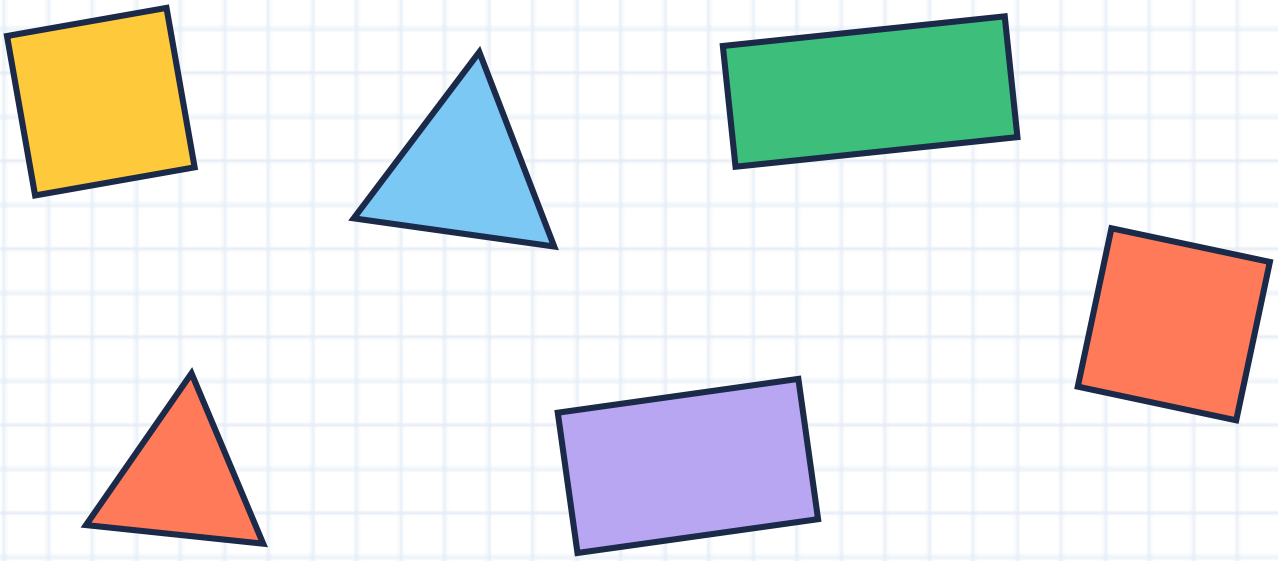


PRIMARY 1 · HOLIDAY PROGRAM

# Math Inquirers

## Notice it. Test it. Explain why.

**Name** _____ **Group** _____

### My Inquirer Rules

1. Notice the pattern.
2. Test it with objects first.
3. Check your answer.
4. Explain why.

# My Inquiry Journal

Your tutor stamps each lesson.

**Lesson 1**
**Count with the Frog**

**Lesson 2**
**Number Bond Inquiry**

**Lesson 3**
**Let's Plan the Fruit Party**

**Lesson 4**
**Pattern Inquiry**

**Lesson 5**
**Mystery Number**

**Lesson 6**
**Coin Combos**

**Lesson 7**
**Shape Safari**

**Lesson 8**
**Logic Lab**

**Lesson 9**
**Measure It!**

**Lesson 10**
**Challenge Day**

Tutor: _____ Program dates: _____

# COUNT WITH THE FROG!

Name _____ Group _____ Date _____

Look for the pattern. Fill in the missing numbers.

## No.1

**Skip counting. Fill in the blanks.**

**a** Counting in 2s: 2, 4, 6, _____, _____, 12, _____, 16

**b** Counting in 5s: 5, 10, _____, 20, _____, 30, _____

**c** Counting in 3s: 3, 6, _____, 12, _____, 18

## No.2

**What is the rule? Fill in the rule and the next two numbers.**

**a** 4, 7, 10, 13, _____, _____ Rule: _____

**b** 30, 27, 24, 21, _____, _____ Rule: _____

**c** 2, 6, 10, 14, _____, _____ Rule: _____

## No.3

**A frog starts at 0 and hops 4 steps each time. Colour the numbers where the frog lands.**

| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

**a** A second frog also starts at 0 but hops 5 steps each time. Mark its landing numbers with a cross (X) above the number line.

**b** Besides 0, which number do BOTH frogs land on?

**Answer:** _____

## No.4

**Ali counts in 2s from 2 up to 50. Mei counts in 5s from 5 up to 50.**

**a** Which numbers will BOTH of them say?

**Answer:** _____

**b** What do you notice about these numbers?

.....

.....

## No.5

**Mei puts stickers in boxes. Box 1 has 2 stickers. Each box after it has 3 more stickers than the box before.**

| Box | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Stickers | 2 | 5 | | | | | | | | |

**a** Fill in the table. How many stickers are in Box 10?

**Answer:** _____

**b** Look at the numbers of stickers. Are they odd or even? What pattern do you see?

.....

.....

# THINK HARDER

Name _____ Group _____ Date _____

*Use objects, draw a picture or make a table. Then write how you know. These pages are for early finishers or for home.*

## Stretch

**No.T1**

**Fill in the missing numbers: 3, 8, ___, 18, ___, 28**

**No.T2**

**Fill in the missing numbers: 40, ___, 28, ___, 16, 10**

**No.T3**

**A pattern starts at 3 and adds 5 each time. What is the 8th number?**

**No.T4**

**A pattern starts at 2 and adds 4 each time. What is the 7th number?**

## Challenge (for strong thinkers)

**No.C1**

**Frog A starts at 3 and jumps 4 each time. Frog B starts at 5 and jumps 3 each time. Which numbers up to 40 do both frogs land on?**

**No.C2**

**1, 2, 4, 7, 11, 16 ... What is the 8th number? Explain how the pattern grows.**

**No.C3**

**Ali counts up from 2 in 3s. Mei counts up from 1 in 4s. Which numbers up to 50 are in both patterns?**

# NUMBER BOND INQUIRY

Name _____ Group _____ Date _____

## No.1

**Number bonds of 10. Fill in the blanks.**

| | | | | |
|---|---|---|---|---|
| 10 = 3 + _____ | 10 = _____ + 6 | 10 = 8 + _____ | 10 = _____ + 5 | 10 = 0 + _____ |

## No.2

**List the pairs that make 10, from the smallest first number.**

**a** 0 + 10, 1 + 9, 2 + _____, 3 + _____, 4 + _____, 5 + _____

**b** When the first number goes up by 1, what happens to the second number?

**c** How many different pairs make 10 if 3 + 7 and 7 + 3 count as the SAME pair?

**Answer:** _____

## No.3

**Use bonds of 10 to help you with bonds of 20.**

**a** 6 + 4 = 10, so 16 + 4 = _____ and 6 + 14 = _____

**b** 7 + 3 = 10, so 17 + 3 = _____ and 7 + 13 = _____

**c** 20 = 12 + _____ 20 = _____ + 9 20 = 15 + _____

## No.4

**The = sign means "is the same as". Fill in the blanks.**

**a** 5 + 3 = 4 + _____ (b) 7 + _____ = 9 + 2

**c** 3 + 8 = _____ + 5 (d) 6 + 6 = 10 + _____

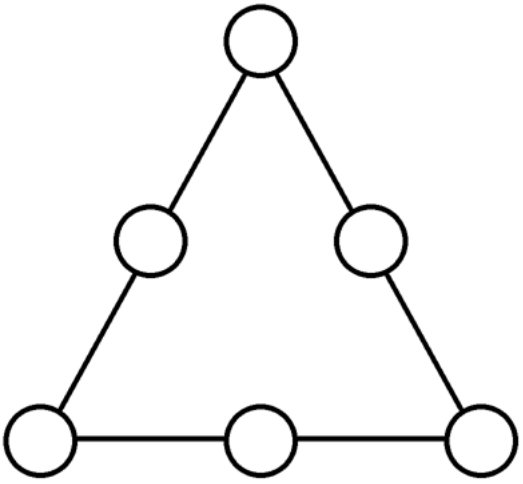
**e** Mei says "4 + 4 = 8 + 1 is true." Is she right? Explain.

.....

.....

## No.5

**Magic triangle. Put the numbers 1, 2, 3, 4, 5 and 6 in the circles so that each side of the triangle adds up to 9.**



**a** Which numbers did you put in the corners? _____

**b** Challenge: now make each side add up to 10. Which numbers are in the corners this time? _____

## No.6

**Two numbers add up to 10. One is 2 more than the other. What are the numbers?**

**Answer:** _____

Challenge: two numbers add up to 12. One is 4 more than the other. What are they?

_____

# THINK HARDER

Name _____ Group _____ Date _____

*Use objects, draw a picture or make a table. Then write how you know. These pages are for early finishers or for home.*

## Stretch

**No.T1**

5 + 4 = 6 + ___

**No.T2**

8 + 2 = 7 + ___

**No.T3**

9 + 6 = ___ + 10

**No.T4**

12 + 3 = 10 + ___

## Challenge (for strong thinkers)

**No.C1**

**Three different numbers from 1 to 7 add up to 10. Find every set of three.**

**No.C2**

**Put 1 to 6 in the magic triangle so each side adds to 11. What goes in the corners and middles?**

**No.C3**

**Ali has 20 stickers. He keeps 9 and shares the rest between Ben and Mei. Ben gets 3 more than Mei. How many does each get?**

# LET'S PLAN THE FRUIT PARTY!

Name _____ Group _____ Date _____

Our teacher wants to buy fruit for a class party. Every child voted for ONE fruit. The picture graph shows the votes.

**Key:** ● = 1 child

| Fruit | Votes |
|---|---|
| Apple | ● ● ● ● ● ● |
| Banana | ● ● ● ● ● |
| Orange | ● ● ● ● ● ● ● ● |
| Grape | ● ● ● |
| Watermelon | ● ● ● ● ● |

## Part A: Read the graph

**No.1**

Which fruit got the most votes?

Answer: _____

**No.2**

Which fruit got the fewest votes?

Answer: _____

**No.3**

How many more children chose orange than grape?

Answer: _____

**No.4**

Two fruits got the same number of votes. Which two fruits are they?

Answer: _____

**No.5**

**How many children voted altogether? (Hint: each child had only 1 vote.)**

**Answer:** _____

**No.6**

**Mei says, "Orange is the best fruit for the whole school!"**

**a** Is Mei right? **Yes / No**

**b** Why do you think so?

.....

.....

**No.7**

**What else should the teacher think about before buying fruit for the party?**

.....

.....

.....

## Part B: Now it's your turn!

Ask your own group to vote. Each child votes for ONE fruit. Make a tally mark for each vote.

| Fruit | Tally | Total |
|---|---|---|
| Apple | | |
| Banana | | |
| Orange | | |
| Grape | | |
| Watermelon | | |

**No.8**

**Which fruit did YOUR group like best? _____ Is it the same as the graph above? Yes / No**

**No.9**

**Why might two groups choose different fruits?**

.....

.....

# THINK HARDER

Name _____ Group _____ Date _____

*Use objects, draw a picture or make a table. Then write how you know. These pages are for early finishers or for home.*

## Stretch

**No.T1**

**1 picture = 2 children. Red has 3 pictures, blue has 5 and green has 4. How many children chose blue?**

**No.T2**

**1 picture = 2 children. Red has 3 pictures, blue has 5 and green has 4. How many children voted altogether?**

**No.T3**

**1 picture = 2 children. Red has 3 pictures and blue has 5. How many more children chose blue than red?**

**No.T4**

**1 picture = 2 children. Six children chose pizza. How many pictures should we draw?**

## Challenge (for strong thinkers)

**No.C1**

1 picture = 3 children. Apple has 4 pictures, banana 2 and orange 5. Then 7 more children join and all choose banana. Which fruit wins now, and by how many votes?

**No.C2**

24 children voted. Apple got 6 votes. Orange got 6 more votes than banana. How many votes did banana and orange get?

**No.C3**

1 picture = 2 children. 20 children voted for three fruits. A has 3 pictures. B has 2 more pictures than A. How many children chose C?

# PATTERN INQUIRY

Name _____ Group _____ Date _____

## No.1

**Look at the shape pattern. Draw the missing shapes in places 8 and 9.**




## No.2

**The pattern keeps going in the same way.**

a What shape will be in place 10? _____

b What shape will be in place 12? _____

c Without drawing, what shape will be in place 30? _____

d How do you know?

.....

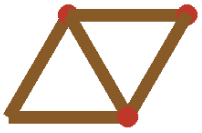
.....

## No.3

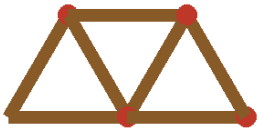
**Ben makes a row of triangles with sticks.**



**1 triangle**



**2 triangles**



**3 triangles**

**a** Count the sticks. Fill in the table. Keep going until you reach 10 triangles.

| Triangles | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Sticks | 3 | 5 | 7 | | | | | | | |

**b** Each time Ben adds 1 more triangle, how many more sticks does he need?

**Answer:** _____

**c** Why does he need only that many more sticks? (Look at the picture.)

.....

.....

**d** How many sticks are needed for 10 triangles?

**Answer:** _____

## No.4

**Challenge! Ben has 15 sticks. How many triangles in a row can he make?**

**Answer:** _____

*Show how you worked it out:*

.....

.....

# THINK HARDER

Name _____ Group _____ Date _____

*Use objects, draw a picture or make a table. Then write how you know. These pages are for early finishers or for home.*

## Stretch

**No.T1**

**Pattern: circle, triangle, triangle, square, circle, triangle, triangle, square ... What is the 14th shape?**

**No.T2**

**Same pattern: how many triangles are in the first 20 shapes?**

**No.T3**

**Same pattern: in which places do the squares come? Write the first five.**

**No.T4**

**Red, blue, blue, red, blue, blue ... What colour is the 20th bead?**

## Challenge (for strong thinkers)

**No.C1**

**Squares in a row use 4, 7, 10 ... sticks. Ben has 50 sticks. How many squares can he make? How many sticks are left?**

**No.C2**

**Beads repeat in a pattern of 5: red, red, blue, blue, green. How many green beads are in the first 47 beads?**

**No.C3**

**Dots: 1, 3, 6, 10, 15 ... What are the next two numbers? What is the 10th number?**

# MYSTERY NUMBER

Name _____ Group _____ Date _____

To find a mystery number, work backwards. Do the opposite steps in the opposite order.

## No.1

**I think of a number. Find my number and write it in the yellow box.**

**a** I add 5. I get 12.

| | | | | |
|---|---|---|---|---|
| **?** | → | + 5 | → | **12** |

**b** I take away 3. I get 9.

| | | | | |
|---|---|---|---|---|
| **?** | → | – 3 | → | **9** |

**c** I add 4. Then I take away 2. I get 10.

| | | | | | | |
|---|---|---|---|---|---|---|
| **?** | → | + 4 | → | – 2 | → | **10** |

## No.2

**A number machine adds the same number every time.**

**a** The machine adds 6. Fill in the table.

| | | | | |
|---|---|---|---|---|
| **IN** | 2 | 5 | 9 | |
| **OUT** | 8 | 11 | | 14 |

**b** A different machine. What does it do to every number?

| | | | | |
|---|---|---|---|---|
| **IN** | 3 | 6 | 10 | 15 |
| **OUT** | 7 | 10 | 14 | |

Rule: _____ If OUT is 30, then IN is _____

**No.3**

**Ali has 8 marbles. Mei has 5 more marbles than Ali. Ben has 6 fewer marbles than Mei.**

**a** How many marbles does Mei have? _____

**b** How many marbles does Ben have? _____

**c** How many more marbles does Mei have than Ben? _____

**d** How many marbles do they have altogether? _____

*Draw a bar for each child to help you.*

**No.4**

**Queue puzzles.**

**a** There are 20 children in a queue. Mei is 7th from the front. How many children are behind her?

**Answer:** _____

**b** Ali is 8th from the front AND 8th from the back. How many children are in the queue? (Careful: Draw it!)

**Answer:** _____

**No.5**

**Make your own mystery number story. Give it to a friend to solve!**

# THINK HARDER

Name _____ Group _____ Date _____

*Use objects, draw a picture or make a table. Then write how you know. These pages are for early finishers or for home.*

## Stretch

**No.T1**

**Ben had some marbles. He gave 5 to Mei, then won 3. Now he has 12. How many did he start with?**

**No.T2**

**Ben had some stickers. He gave away 6, got 4, then gave away 3. Now he has 10. How many did he have at first?**

**No.T3**

**A number machine adds the same number each time. IN 4 gives OUT 9. What is the rule? What is OUT when IN is 12? What is IN when OUT is 20?**

**No.T4**

**I think of a number. I add 4, then I take away 7. I get 8. What is my number?**

## Challenge (for strong thinkers)

**No.C1**

I think of a number. I add 7, take away 3, add 5, then take away 8. I get 14. What is my number?

**No.C2**

Ali and Ben have 20 marbles together. Ali has 4 more than Ben. How many marbles does each have?

**No.C3**

A snail climbs 3 cm each day and slides back 1 cm each night. It starts at the bottom of a 9 cm wall. On which day does it reach the top?

# COIN COMBOS

Name _____ Group _____ Date _____

We have 5¢, 10¢, 20¢ and 50¢ coins. You can use the same kind of coin more than once.

**No.1**

**Show different ways to make exactly 20¢ using only 20¢, 10¢ and 5¢ coins. Write how many of each coin in the table.**

| Way | Number of 20¢ | Number of 10¢ | Number of 5¢ |
|---|---|---|---|
| A | | | |
| B | | | |
| C | | | |
| D | | | |
| E | | | |

a Which way uses the FEWEST coins?

**Answer:** _____

b Which way uses the MOST coins?

**Answer:** _____

c How do you know you have found ALL the ways?

.....

.....

## No.2

**Challenge! Find ALL the ways to make 30¢ using 20¢, 10¢ and 5¢ coins. Write each way like this: 20¢ + 5¢ + 5¢**

## No.3

**Mei wants to pay exactly 85¢ using 5¢, 10¢, 20¢ and 50¢ coins.**

**a** What is the smallest number of coins she can use? Which coins are they?

**Answer:** _____

**b** Find a different way that uses exactly 6 coins.

**Answer:** _____

## No.4

**Can Ben make 35¢ using only 2 coins? Explain your answer.**

## No.5

**Dollars! We also have $1 coins and $2, $5, $10 and $50 notes.**

**a** A toy costs $17. Pay exactly with the fewest notes and coins.

**Answer:** _____

**b** Show 3 different ways to pay exactly $20.

**c** Mei has one $50 note and two $10 notes. A bicycle costs $65 and a helmet costs $8. Does she have enough money for both? Explain.

# THINK HARDER

Name _____ Group _____ Date _____

*Use objects, draw a picture or make a table. Then write how you know. These pages are for early finishers or for home.*

## Stretch

### No.T1

**A bag costs $34. Ali pays with a $50 note. How much change does he get?**

### No.T2

**Make 70¢ using exactly 4 coins from 5¢, 10¢, 20¢ and 50¢. Find every set.**

### No.T3

**Sam has only 10¢ and 20¢ coins. He pays exactly 90¢ using 6 coins. How many of each coin does he use?**

### No.T4

**What is the fewest number of coins to make 45¢?**

## Challenge (for strong thinkers)

**No.C1**

Ali pays 55¢ with exactly 5 coins. He uses at least one 5¢, one 10¢ and one 20¢ coin. Which coins does he use?

**No.C2**

You have one 5¢, one 10¢, one 20¢ and one 50¢ coin. How many different amounts can you make using any of them?

**No.C3**

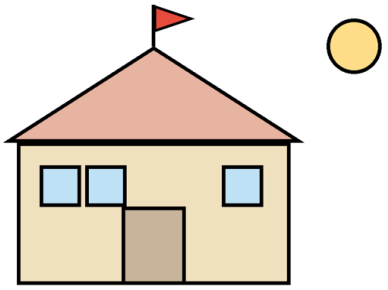
A sweet costs 15¢ and a biscuit costs 20¢. Ben spends exactly 90¢ and buys some of each. How many sweets and biscuits does he buy?

# SHAPE SAFARI

Name _____ Group _____ Date _____

## No.1

**Count the shapes in the house picture.**



| Circle | Triangle | Square | Rectangle |
|---|---|---|---|
| | | | |

a Which shape is there the most of? _____ (b) How many shapes altogether? _____

## No.2

**Sides and corners. Fill in the table.**

| Shape | Straight sides | Corners |
|---|---|---|
| Circle | 0 | 0 |
| Triangle | 3 | 3 |
| Square | | |
| Pentagon | | |
| Hexagon | | |
| Octagon | | |

a What do you notice about the number of sides and corners?

_____

b A shape has 7 sides. How many corners does it have? _____

## No.3

**Who am I?**

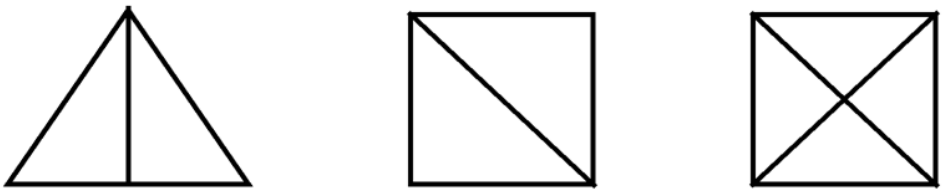
**a** I have 4 sides, all the same length, and 4 square corners. _____

**b** I have no corners and no straight sides. _____

**c** I have 6 sides. _____

## No.4

**How many triangles can you find in each picture? Count the small ones first, then the bigger ones.**



A B C

**A:** _____ **B:** _____ **C:** _____

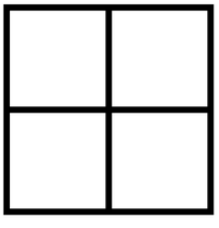
## No.5

**Two triangles of the same size are put together along one side. Draw one shape you can make. What is it called?**

**Shape name:** _____

## No.6

**How many squares can you find in this picture? Count the small ones first, then the bigger one.**



**Number of squares:** _____

# THINK HARDER

Name _____ Group _____ Date _____

*Use objects, draw a picture or make a table. Then write how you know. These pages are for early finishers or for home.*

## Stretch

### No.T1

**Fold a square along a diagonal and cut. Which shapes do you get?**

### No.T2

**Two same-size triangles can make a square. Can they also make a bigger triangle? Draw how.**

### No.T3

**How many squares are in a grid 3 squares long and 2 squares high?**

### No.T4

**Join 4 same-size squares, whole sides touching. How many different shapes can you make? (Turns and flips are the same shape.)**

## Challenge (for strong thinkers)

**No.C1**

Three squares are joined in a row to make a long rectangle. How many rectangles can you find altogether? (A square counts as a rectangle.)

**No.C2**

A triangle has 4 lines drawn from the top corner to the bottom. How many triangles are there? Follow the pattern 3, 6, 10 ...

**No.C3**

A square is cut into 4 equal squares. One of the small squares is cut into 4 even smaller squares. How many squares can you find altogether?

# LOGIC LAB

Name _____ Group _____ Date _____

## No.1

**Put them in order using the clues.**

a Ali is taller than Ben. Ben is taller than Mei. Tallest: _____ Shortest: _____

b The cat is heavier than the dog. The hen is lighter than the dog. Heaviest: _____ Lightest: _____

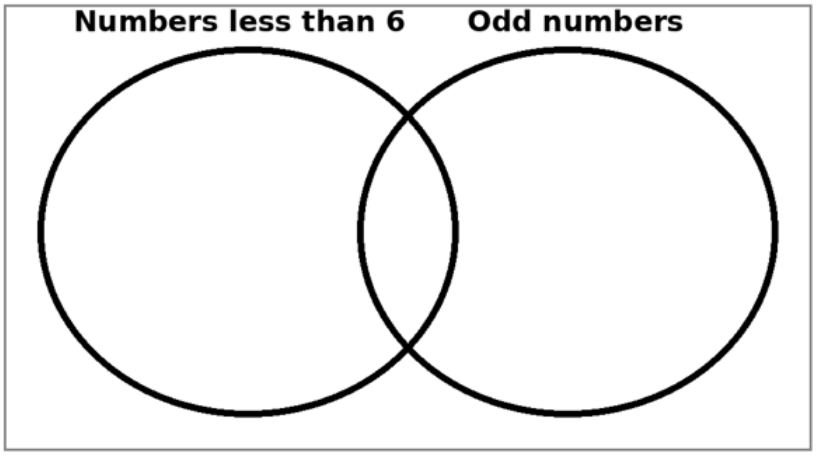
c The red ribbon is longer than the blue one.
The green one is shorter than the blue one.
The yellow one is longer than the red one.
Write them from longest to shortest:

_____, _____, _____, _____

## No.2

**Write each number from 1 to 10 in the right place in the two-loop picture.**

*Odd numbers end in 1, 3, 5, 7 or 9. Even numbers end in 0, 2, 4, 6 or 8.*




a Numbers that are in BOTH circles: _____

b Why do those numbers belong in the middle?

## No.3

**Who has which pet? Ali, Mei and Ben each have a different pet: a cat, a dog or a fish.**

Clue 1: Ali does not have a cat or a dog.

Clue 2: Mei does not have a dog.

*Put a tick where it fits and a cross where it cannot.*

| | Cat | Dog | Fish |
|---|---|---|---|
| **Ali** | | | |
| **Mei** | | | |
| **Ben** | | | |

**Ali has the _____. Mei has the _____. Ben has the _____.**

## No.4

**Four children stand in a line: Ali, Ben, Mei and Sam.**

Mei is first. Ali is in front of Ben. Ben is in front of Sam. Where does each child stand?

| 1st | 2nd | 3rd | 4th |
|---|---|---|---|
| | | | |

## No.5

**Number riddles. Who am I?**

**a** I am between 10 and 20. I am odd. When you count in 5s, you say me. _____

**b** I am between 20 and 40. When you count in 2s you say me. When you count in 5s you say me too. _____

**c** I have 2 digits. My digits add up to 5. I am less than 20. _____

## No.6

**Which one does NOT belong? 3, 6, 9, 11**

Explain why. Then try to find a second reason!

# THINK HARDER

Name _____ Group _____ Date _____

*Use objects, draw a picture or make a table. Then write how you know. These pages are for early finishers or for home.*

## Stretch

**No.T1**

**In a race, Sam is ahead of Tia. Tia is ahead of Raj. Mei is ahead of Sam. Who is third?**

**No.T2**

**Ana, Bo and Cy each have a different toy: a ball, a kite or a yo-yo. Bo does not have the kite. Cy has the ball. Who has which toy?**

**No.T3**

**Numbers 1 to 12. Which do we say when counting in 2s? Which when counting in 3s? Which in both?**

**No.T4**

**Numbers 1 to 12. Which do we NOT say when counting in 2s or when counting in 3s?**

## Challenge (for strong thinkers)

**No.C1**

**Five friends run a race. D is first. E is last. B finishes directly behind A. C is not second. In what order do they finish?**

**No.C2**

**A secret number is between 10 and 30. It is even, and you say it when counting in 3s. It is 2 more than a number you say when counting in 5s. What is it?**

**No.C3**

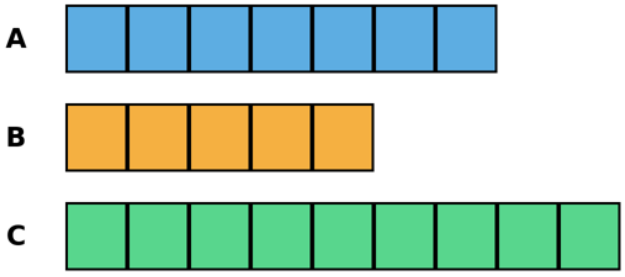
**Numbers 1 to 20. How many do we say only when counting in 2s, only in 3s, in both, and in neither?**

# MEASURE IT!

Name _____ Group _____ Date _____

## No.1

**Three ribbons are made of equal squares.**




a Which ribbon is the longest? _____ (b) Which is the shortest? _____

c How many squares longer is ribbon A than ribbon B? _____

d Ribbon A and ribbon B are put end to end. How many squares long is that? _____

e Ribbon C is cut shorter by 2 squares. Which ribbon is it now the same length as? _____

## No.2

**Measure with cubes. Use the paper cube strip from the back of the workbook (or real cubes). Guess first, then measure. Write how far off your guess was.**

| Object | My guess | Measured | Difference (guess and measure) |
|---|---|---|---|
| A pencil | | | |
| A book | | | |
| A shoe | | | |
| My choice: _____ | | | |

a Why must every cube be the same size?

.....

**No.3**

**Ali measures a desk with his hand span. It is 6 spans long. Mei measures the same desk with her hand span. It is 8 spans long.**

**a** Whose hand span is longer? _____

**b** How do you know?

.....

.....

**No.4**

**One paper clip is as long as 2 erasers.**

**a** A desk is 5 paper clips long. How many erasers long is it? _____

**b** A pencil is 12 erasers long. How many paper clips long is it? _____

**No.5**

**Mei's pencil is 8 cubes long. Ali's pencil is 3 cubes longer than Mei's. Ben's pencil is 2 cubes shorter than Mei's.**

**a** How long is Ali's pencil? _____ cubes

**b** How long is Ben's pencil? _____ cubes

**c** If all three pencils are put end to end, how many cubes long are they? _____ cubes

**No.6**

**Use your ruler.**

**a** A pencil lies from the 0 cm mark to the 12 cm mark. How long is it? _____ cm

**b** A crayon lies from the 3 cm mark to the 8 cm mark. How long is it? _____ cm

**c** Measure your own pencil with your ruler. _____ cm. Is it longer or shorter than 10 cm? _____

# THINK HARDER

Name _____ Group _____ Date _____

*Use objects, draw a picture or make a table. Then write how you know. These pages are for early finishers or for home.*

## Stretch

**No.T1**

**A caterpillar is 9 cubes long. It grows 2 cubes each day. How long is it after 3 days?**

**No.T2**

**A string is 15 cubes long. A rope is 8 cubes longer. How long is the rope?**

**No.T3**

**A ribbon lies from the 3 cm mark to the 11 cm mark. How long is it?**

**No.T4**

**A pencil is 3 erasers long. A ruler is 2 pencils long. A desk is 4 rulers long. How many erasers long is the desk?**

## Challenge (for strong thinkers)

**No.C1**

A stick is 12 cubes long. Ali measures a desk by laying the stick end to end 4 times and then 5 more cubes. How long is the desk in cubes?

**No.C2**

Ali's pencil is 9 cubes. Mei's is 2 cubes shorter than Ali's. Ben's is 3 cubes longer than Mei's. How long are all three pencils together?

**No.C3**

A ribbon 30 cubes long is cut into three pieces. The first is 4 cubes longer than the second. The third is 4 cubes shorter than the second. How long is each piece?

# CHALLENGE DAY!

Name _____ Group _____ Date _____

*Show how you know. Remember the inquirer rules!*

## No.1

**Pattern: 3, 8, 13, 18, _____, _____ Rule: _____**

## No.2

**Two numbers add up to 12. One is 2 more than the other. What are they? _____ and _____**

## No.3

**The graph shows pets in a street.**

| | |
|---|---|
| **Dog** | ● ● ● ● |
| **Cat** | ● ● ● ● ● ● ● |
| **Fish** | ● ● ● |
| **Bird** | ● ● ● ● ● |

● = 1 pet

**a** How many more cats than birds? _____

**b** How many pets are there altogether? _____

**c** Which two kinds of pet together make exactly 10? _____

## No.4

**You have one 20¢ coin, two 10¢ coins and two 5¢ coins.**

**a** How much money do you have? _____

**b** Which of your coins make exactly 35¢? _____

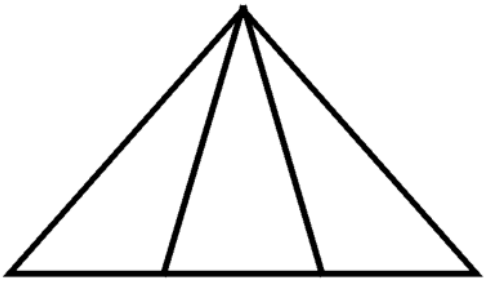
**No.5**

**Shapes**

**a** I have 4 sides, all the same length, and 4 square corners. What am I?

_____

**b** How many triangles are in this picture? _____



**No.6**

**Ali is older than Ben. Mei is younger than Ben. Who is the youngest?**

_____

**No.7**

**I think of a number. I add 9. Then I take away 4. I get 10. What is my number? _____**

**No.8**

**A pencil is 6 cubes long. A pen is 4 cubes longer than the pencil.**

**a** How long is the pen? _____ cubes

**b** Put the pen and the pencil end to end. How long are they together? _____ cubes

**No.9**

**Make a challenge puzzle for a friend. Write the puzzle and the answer.**

.....

.....

.....

.....

The puzzle I liked best today was _____ because _____

# THINK HARDER

Name _____ Group _____ Date _____

*Use objects, draw a picture or make a table. Then write how you know. These pages are for early finishers or for home.*

## Stretch

**No.T1**

**A pattern starts at 5 and adds 3 each time. What is the 10th number?**

**No.T2**

**Ali has 12 stickers. He gives 5 to Mei and then gets 3 more. How many does he have now?**

**No.T3**

**Which three coins from 5¢, 10¢, 20¢ and 50¢ make 45¢?**

**No.T4**

**A ribbon is 12 cubes. A string is 5 cubes shorter. How long is the string?**

## Challenge (for strong thinkers)

**No.C1**

Frog A jumps in 3s from 0 and frog B jumps in 5s from 0. Which numbers up to 60 do both land on?

**No.C2**

Ali has some stickers. Ben has 4 more than Ali. Mei has 3 fewer than Ben. Together they have 29. How many does each have?

**No.C3**

Make 70¢ using exactly 4 coins from 5¢, 10¢, 20¢ and 50¢. How many ways are there?

# Cut-outs

For Lesson 6 (Coin Combos) and Lesson 9 (Measure It!). Cut along the dashed lines and keep the pieces in an envelope.

## Play coins (seven 5¢, seven 10¢, six 20¢, five 50¢)

5¢ 5¢ 5¢ 5¢ 5¢ 5¢ 5¢

10¢ 10¢ 10¢ 10¢ 10¢ 10¢ 10¢

20¢ 20¢ 20¢ 20¢ 20¢ 20¢

50¢ 50¢ 50¢ 50¢ 50¢

## Paper cube strips (every square is the same size)

### Strip A

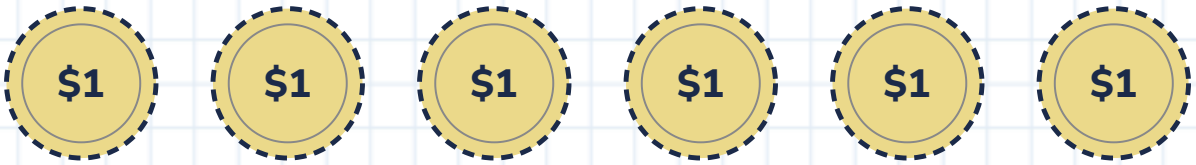
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

### Strip B

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

# Play notes and $1 coins

For Lesson 6 (Coin Combos). Six $1 coins, four $2, four $5, four $10 and two $50 notes.

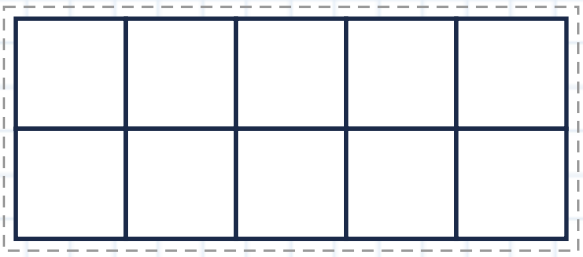




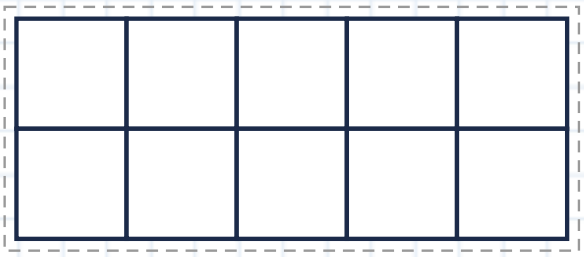

# Inquirer Kit

Nothing needs to be bought. Print, cut out and keep everything in one envelope.

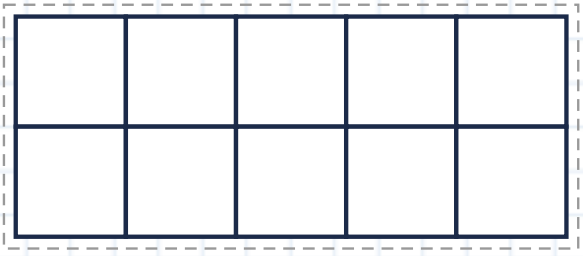
**Ten-frame 1**



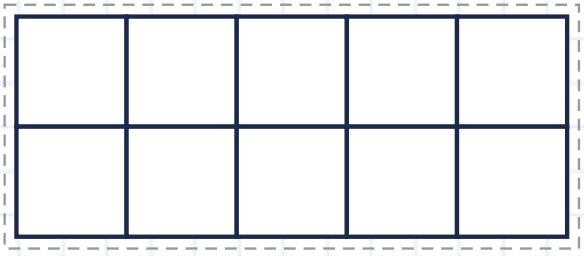
**Ten-frame 2**



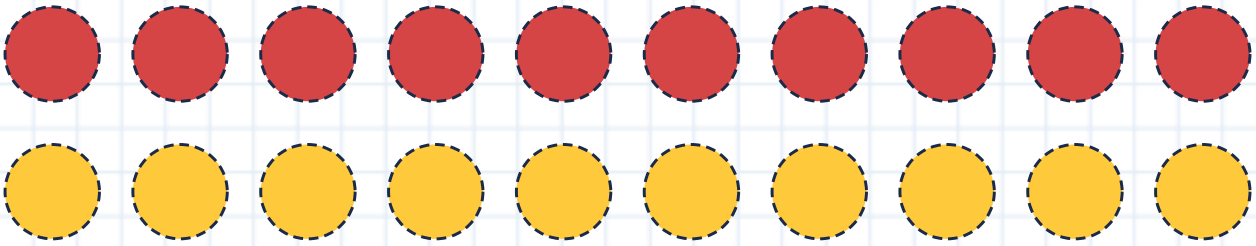
**Ten-frame 3**



**Ten-frame 4**



**Counters: cut out all 20 (10 red, 10 yellow)**



Cut along the dashed lines.

# Number strips and sorting tiles

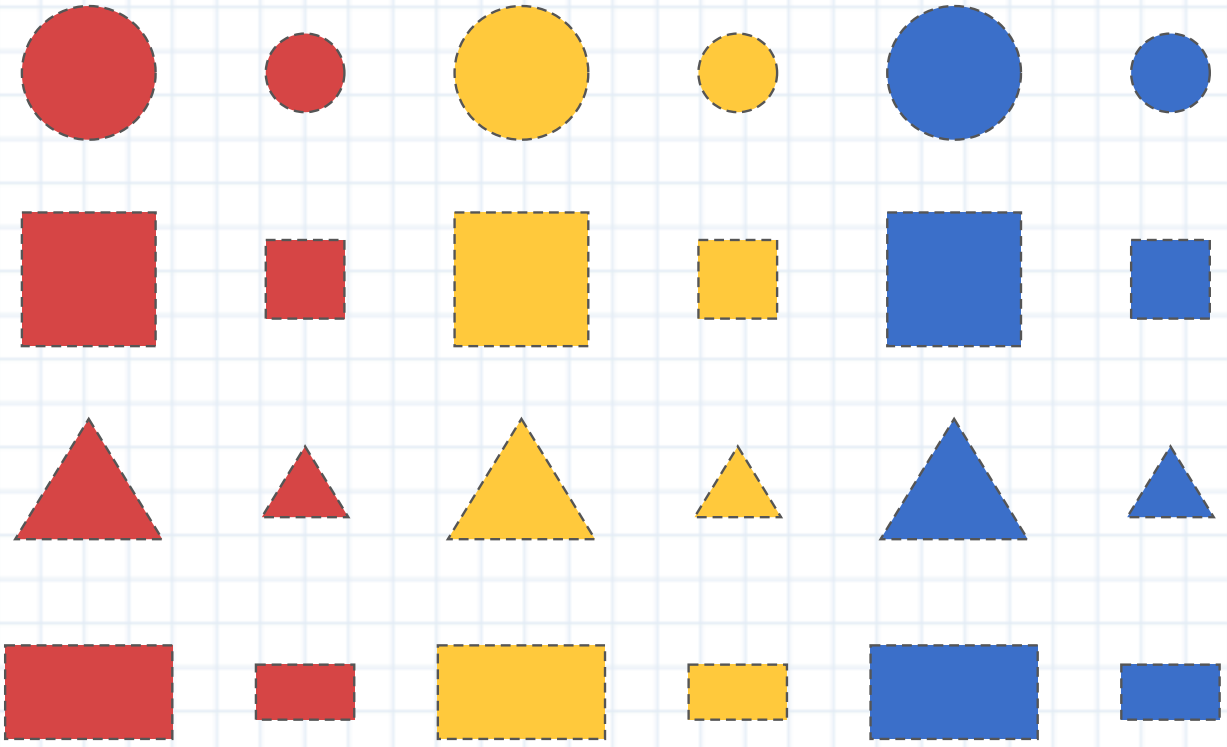
## Number strips 0 to 20 (cut out; use as a number line)

| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

## Shape and sorting tiles (cut out all 24: 4 shapes, 3 colours, 2 sizes)



Shapes: circle, square, triangle, rectangle. Colours: red, yellow, blue. Sizes: big, small.

# Puzzle cards

For Lesson 10 (Challenge Day). Cut along the dashed lines. Your tutor tells you which card to start with. Show how you know on a whiteboard or on paper.

### 1. COINS

Make 45¢ with exactly 3 coins from 5¢, 10¢, 20¢ and 50¢.

### 2. PATTERNS

5, 8, 11, 14 … What is the 10th number? Show how you know.

### 3. LOGIC

Mei is taller than Ben. Ben is taller than Sam. Ali is shorter than Sam. Put them in order, tallest first.

### 4. MEASURING

A pencil is 6 cubes long. A pen is 4 cubes longer than the pencil. Put them end to end. How many cubes long is that?

# CERTIFICATE OF COMPLETION

## Math Inquirers Holiday Program

Primary 1 Holiday Program

This certifies that

.....

completed all 10 lessons, solved puzzles, spotted patterns and explained their thinking.

## Notice it. Test it. Explain why.

Tutor: _____ Date: _____